



PROJECT 12

EMPLOYEE SCORE

Linear Transformation

PRESENTED BY MUNAWAR

BACKGROUND

This analysis of Employee Performance data aims to delve deeper into the factors that influence individual evaluation scores (emp-scores). Because the HR department needs to present these regression models to upper management, the insights must be easily explainable. Specifically, HR is interested in how experience level, work hours, and degree types drive variations in employee scores.

To improve model interpretation without affecting predictive performance can applies linear standardization. Standardization can be done in various way:

- Scaling predictors using reasonable unit scales.
- Centering variables around their mean.
- Standardizing using Z-scores.

Additionally, to enhance the interpretation of models with interaction terms, we utilize centering and standardization. For models including binary predictors, we apply a two-standard-deviation scaling approach to keep coefficients comparable.

In this project used Standardizing using Z-scores and Centering variables

EXPLORATORY DATA ANALYSIS

Statistik Descriptive

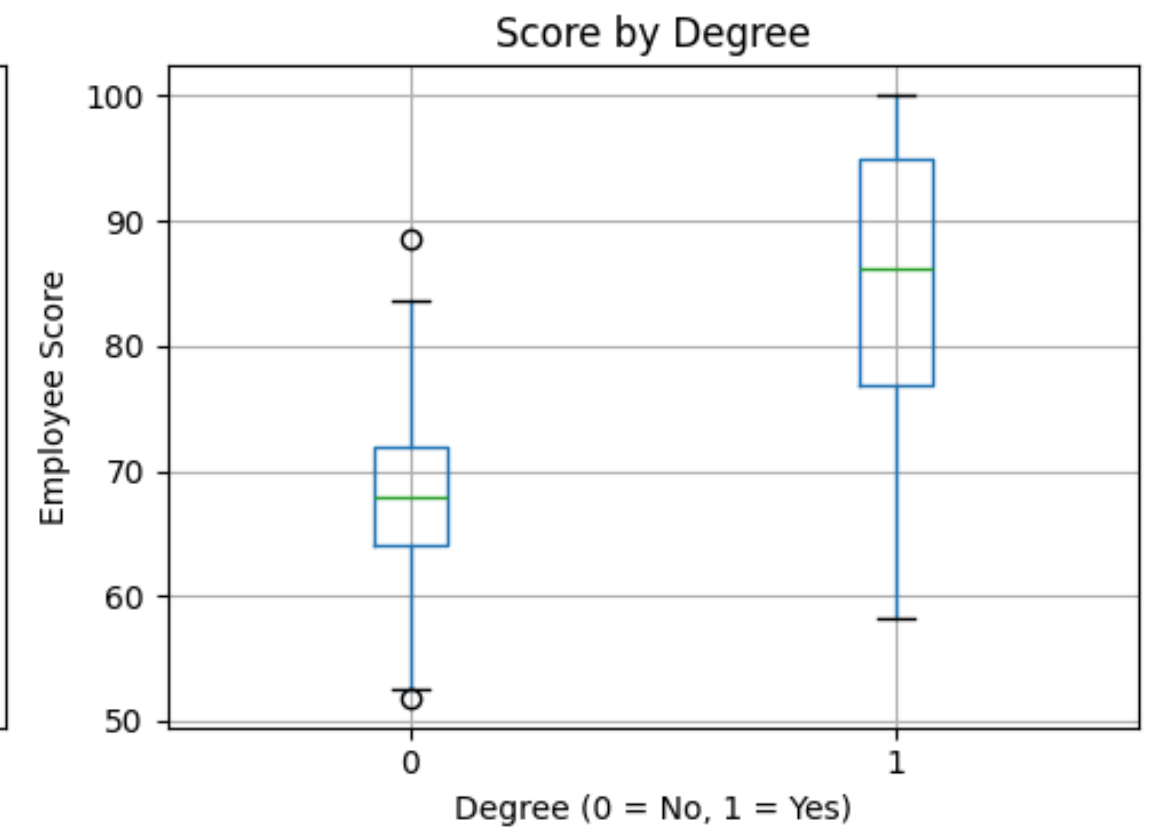
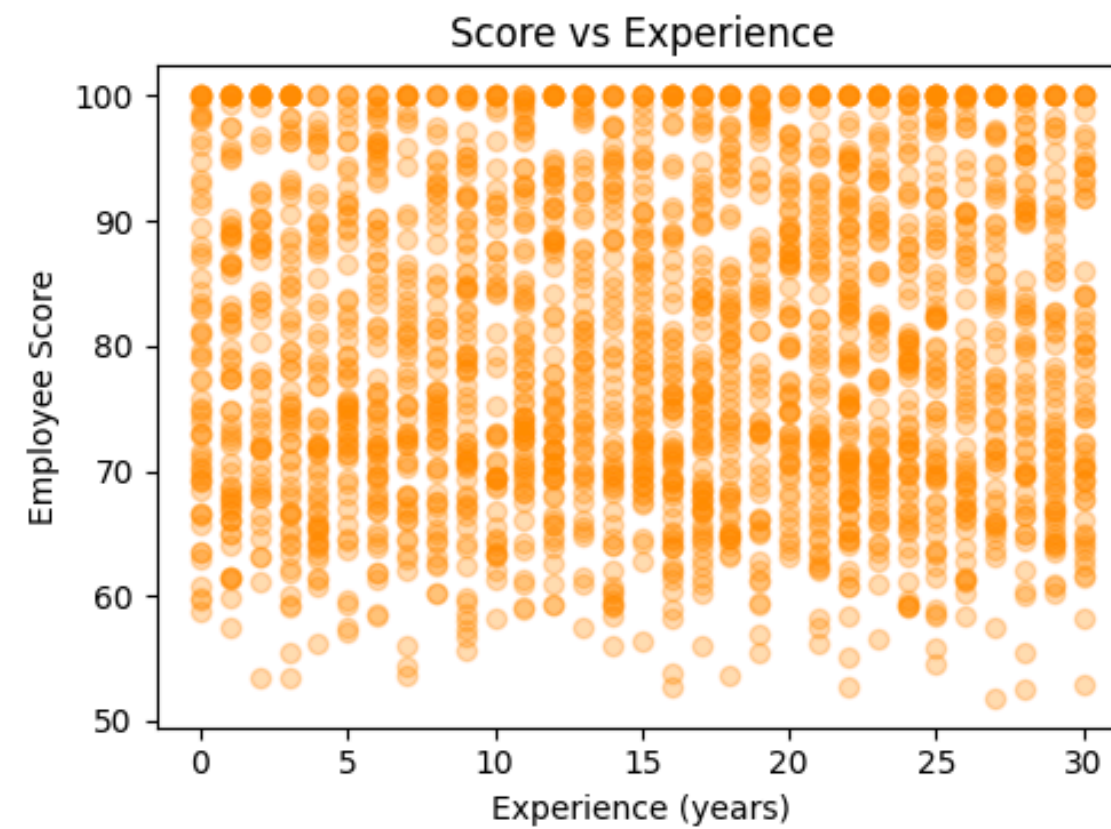
	emp_score	emp_degree	emp_experience	emp_hours	emp_age
count	2160.000000	2160.000000	2160.000000	2160.000000	2160.000000
mean	79.867454	0.683333	15.259259	7.995185	38.984722
std	12.541678	0.465284	8.805889	2.337024	12.452159
min	51.900000	0.000000	0.000000	4.000000	18.000000
25%	69.700000	0.000000	8.000000	5.900000	28.000000
50%	78.300000	1.000000	16.000000	8.100000	39.000000
75%	90.725000	1.000000	23.000000	10.000000	50.000000
max	100.000000	1.000000	30.000000	12.000000	60.000000

emp_score	
emp_degree	
0	67.815205
1	85.452642

- The average score the employee is about 79.
- The average employee experience is about $\approx$ 15 years.
- The average employee work hours is approximately 8 hours per day, with a range of 4 to 12 hours.

- Employees with a degree score on average 17.6 points higher than those without

EXPLORATORY DATA ANALYSIS



BASE LINEAR REGRESSION MODEL

$$emp_{score} = 43.08 + 0.012emp_{experience} + 3.10emp_{hours} + 17.17emp_{degree}$$

Parameter	Estimate	Std Error	Interpretation
Intercept	43.08	0.5489	Predicted score when all = 0 (not meaningful)
emp_experience	0.012	0.014	Std err > coef — high uncertainty, weak predictor
emp_hours	3.10	0.056	+1 hr/day → +3.11 score points
emp_degree	17.17	0.2823	Degree vs no degree → +17.18 score points

R² = 0.763

76.3% variance explained

Intercept (43.09) is not meaningful — no employee has 0 hrs, 0 experience, and no degree. We can improve this.

STEP 1 – STANDARIZATION DEGREE (2-SD SCALING)

$$emp_{score} = 43.08 + 0.012emp_{experience} + 3.10emp_{hours} + 17.17emp_{degree}$$

BEFORE (original emp_degree)	AFTER (z2_degree)
Intercept: 43.09 Meaning: predicted score for an employee with NO degree, 0 hrs, and 0 experience. → Not a useful real-world value.	Intercept: 54.82 Meaning: predicted score for an employee with AVERAGE degree level, 0 hrs, and 0 exp. → More informative baseline.

Intercept now represents the average degree employee — meaningful for upper management reporting

STEP 2 – CENTER CONTINUOUS PREDICTORS

$$emp_{score} = 67.25 + 0.012emp_{experience(c)} + 3.10emp_{hours(c)} + 15.98emp_{degree(c)}$$

emp_hours_centered

emp_hours – 4 (min value)

emp_experience_centered

emp_experience – 0 (min value)

Final Coefficient Interpretation

Parameter	Estimate	Interpretation
Intercept	43.08	Score for emp at min hrs (4), 0 exp, avg degree
exp_centered (1 yr)	0.012	Same hrs & degree: +0.012 pts per extra year of exp
hrs_centered (1 hr)	3.10	Same exp & degree: +3.11 pts per extra hr/day
z2_degree	17.17	Same exp & hrs: degree vs no degree → +15.98 pts

CONCLUSION

1

Model Explains 76% of Score Variance

The combination of hours, degree, and experience is a strong predictor of performance.

2

Degree Has the Largest Impact (+ 15.98 pts)

Employees with a degree consistently score ~16 points higher, all else being equal.

3

Work Hours Significantly Drive Performance

Each extra hour per day adds 3.11 score points — a meaningful and actionable lever.

4

Experience is Statistically Weak

High standard error ($0.015 > \text{coef } 0.012$) suggests experience alone is not conclusive.

**THANK YOU FOR
YOUR TIME**



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DATA ANALYST PRESENTATION
